



Rapid communication

Rapid estimation of Brilliant Blue concentrations in soil by vis diffuse reflectance spectroscopy

Christina Bogner^{a,*}, Iris Schmiedinger^b, Bernd Huwe^b^a Ecological Modelling, BayCEER, University of Bayreuth, Dr.-Hans-Frisch-Straße 1–3, 95448 Bayreuth, Germany^b Soil Physics Group, BayCEER, University of Bayreuth, 95440 Bayreuth, Germany

ARTICLE INFO

Article history:

Received 31 March 2010

Received in revised form 17 March 2011

Accepted 4 April 2011

Available online 28 May 2011

Keywords:

Vadose zone

Dye tracers

Reflectance spectra

Calibration

Statistical analysis

ABSTRACT

Brilliant Blue is often used to trace water movement in soils. Its concentration in soil samples is usually determined by extraction — a laborious procedure with varying accuracy. In this feasibility study, we show that Brilliant Blue can be estimated directly by visible (vis) diffuse reflectance spectroscopy (DRS). We build a PLSR (partial least squares regression) model for the concentration range of 0.1 to 15 mg Brilliant Blue per g soil with an RMSE (root mean square error) of 1.35 mg g^{-1} and an R^2_{adj} of 0.88. However, as the method is based on visible spectra, prediction accuracy can be altered by variations in soil colour between calibration and prediction data sets. Therefore, we recommend a site specific calibration based on typical soil colours and an expected range of Brilliant Blue concentrations.

© 2011 Elsevier B.V. All rights reserved.

1. Introduction

Water flow and solute transport in soils are complex phenomena. Usually, water chooses certain preferred pathways through the soil, e.g. along root channels, cracks or textural boundaries and bypasses a large portion of the soil matrix (e.g. Flury et al., 1994). This type of flow is called preferential flow and is hardly predictable. Often visualization of preferential flow in tracer studies is the best way to study it. Brilliant Blue FCF ($\text{C}_{37}\text{H}_{34}\text{N}_2\text{Na}_2\text{O}_9\text{S}_3$, molar mass 792.9 g mol^{-1}) is an artificial organic dye frequently used in soil hydrology to trace water movement in soils and to identify preferential flow paths. In a tracer study, an aqueous solution of the dye is evenly applied at the soil surface, often by an automated sprinkler. Then vertical soil profiles are excavated to uncover infiltration patterns of the stained water. These patterns are photographed and the soil is sampled to determine the content of Brilliant Blue. Linking the concentration of the dye to the color information of the image allows to determine concentration maps of Brilliant Blue to understand water and solute transport in soils (Forrer et al., 2000).

Preferential flow can affect soil chemistry and biology. Several authors identified preferential flow paths in Brilliant Blue tracer studies and analysed the content of total and organic C and N (Bundt et al., 2001a,b; Hagedorn et al., 1999). They showed that soil matrix and preferential flow paths differed in their chemical and biological properties. The sorption of the dye on soil particles during infiltration

of the tracer solution changes the C and N contents of the soil. Therefore, the concentration of Brilliant Blue in the soil has to be determined and the additional amounts of C and N due to the dye have to be subtracted prior to interpretation of soil chemical properties.

Consequently, the estimation of Brilliant Blue concentrations in soil is an important step in the analysis of tracer experiments. Usually, the quantity of the dye is determined by extraction with a water acetone solution or a 0.5 M K_2SO_4 (Bundt et al., 2001b). This is a laborious procedure with changing accuracy due to varying mass recovery (Forrer et al., 2000). The dye cannot be extracted completely and the residual concentration remaining in the soil has to be estimated. We propose to determine the content of Brilliant Blue by visible diffuse reflectance spectroscopy (vis DRS) directly on soil samples without extraction. This work is a feasibility study with the focus on a local calibration procedure.

2. Materials and methods

2.1. Soil samples

We took soil material from one profile on two study sites: an agricultural field at the experimental site of INRA Avignon, France ($43^\circ 54' 58.76''\text{N}$, $4^\circ 52' 54.81''\text{E}$), and in a Norway spruce forest in the Fichtelgebirge in south-eastern Germany ($50^\circ 08'\text{N}$, $11^\circ 52'\text{E}$). The soil at the agricultural site is a highly structured Calcisol (IUSS Working Group WRB, 2007) with a homogeneously greyish yellow colour (Oyama & Takehara, 1999) throughout the profile. We took bulk samples above the plough pan in 0–30 cm and below in 30–60 cm and

* Corresponding author. Tel.: +49 921 555655; fax: +49 921 555799.

E-mail address: christina.bogner@web.de (C. Bogner).

Table 1
Soil characteristics.

Site	Horizon or depth (cm)	Sand (%)	Silt (%)	Clay (%)	Colour ^a	Colour name ^a
I ^b	0–30	5.18	48.35	46.47	2.5Y 6/2	Greyish yellow
	30–60	4.94	48.14	46.93	2.5Y 6/2	Greyish yellow
	60–100	15.53	47.63	36.85	2.5Y 6/2	Greyish yellow
II ^c	Ea	43.16	48.47	8.37	10YR 5/2	Greyish yellow brown
	Bsh	38.63	51.48	9.90	10YR 4/2	Greyish yellow brown
	Bs	33.94	50.38	15.69	10YR 4/4	Brown
	Bw	31.61	54.53	13.87	2.5Y 7/4	Light yellow
	Bw/C	36.62	49.71	13.67	2.5Y 7/4	Light yellow

^a Determined on dry samples according to Oyama and Takehara (1999).^b Agricultural site.^c Forest site.

60–100 cm. The forest soil at the second site is a Podzol (IUSS Working Group WRB, 2007) with five different horizons: Ea, Bsh, Bs, Bw, Bw/C. The soil colour varies considerably between horizons from greyish yellow brown to light yellow and we took bulk samples per horizon. Table 1 summarizes the distribution of the soil fine fraction (soil particles <2 mm) and the soil colour.

2.2. Sample preparation and measurements

We prepared 40 calibration samples from the agricultural site (calibration set I) and 65 from the forest site (calibration set II) by mixing 5 g of sieved soil (<2 mm) with 5 ml of Brilliant Blue solution with different concentrations. The final concentration of the dye ranged from about 0.1 to 15 mg g⁻¹ soil for calibration set I and from about 0.1 to 10 mg g⁻¹ soil for calibration set II. The mean concentration step equaled 0.4 mg g⁻¹ and 0.2 mg g⁻¹ for calibration set I and II, respectively. After mixing the soil with the Brilliant Blue solution the samples were dried at 40 °C and ground. Two validation data sets of 20 samples each from the agricultural site (validation set I) and from the forest site (validation set II) were prepared in the same way. Their concentration ranged from about 0.1 mg g⁻¹ to 14 mg g⁻¹ with a mean concentration step of 0.7 mg g⁻¹ and from about 0.3 mg g⁻¹ to 9 mg g⁻¹ with a mean concentration step of 0.5 mg g⁻¹ for the validation set I and II, respectively. These validation data sets were not included in the calibration procedure.

Visible diffuse reflectance spectra of the soil samples were collected using a Cricket accessory (Harrick Scientific Products, Inc., Pleasantville, NY, USA) installed in a Cary 100 Conc UV–vis spectrometer (Varian, Inc., Santa Clara, CA, USA). An aliquot of a ground sample was scanned between 400 to 700 nm in 1 nm steps (wavelength accuracy of the spectrometer equals 0.02 nm at 656.1 nm). To improve accuracy an average of 10 readings per step was recorded. Then, a baseline correction procedure was applied:

$$S = \frac{\text{Raw spectrum} - 0\% \text{Transmission}}{100\% \text{Transmission} - 0\% \text{Transmission}}$$

where *S* is the corrected spectrum, 0%Transmission is the zero reference baseline collected with the sample beam covered and the 100%Transmission is the spectrum collected on spectralon (100% reflectance reference). Spectra were recorded in reflectance units (*R*) and transformed to absorbance units, i.e. log(1/*R*).

2.3. Partial least squares regression

In order to estimate the content of Brilliant Blue in soil samples, we need a model relating soil spectra to the concentration of the dye. Partial least squares regression is a multivariate linear regression technique particularly suitable when predicting a dependent variable

from a large number of independent (possibly collinear) variables. We used the software ParLes (Viscarra Rossel, 2008) that allows multivariate calibration and prediction based on partial least squares regression combined with bootstrap aggregation (bagging-PLSR). For an introduction to PLSR the reader is referred to Geladi and Kowalski (Geladi & Kowalski, 1986). During bagging, the calibration data set is repeatedly sampled with replacement and a PLSR model is calculated for each of these subsamples. Using these models, the independent data are predicted and a mean predictor with confidence intervals can be derived. A tutorial on bootstrapping for chemical application can be found in Wehrens et al. (2000). Viscarra Rossel (2007) showed that bagging improved the robustness of PLSR, was less susceptible to overfitting and led to more accurate models while providing a measure for model uncertainty.

Prior to calibration, soil spectra were mean centred and smoothed with a wavelet filter. To choose the number of factors to use in the PLSR model, a leave-one-out cross validation was performed. The accuracy of the cross validation is measured by the root mean square error (RMSE):

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2}$$

where *N* is the number of samples, *y_i* the observed value, *ŷ_i* the predicted one. The optimal number of factors, i.e. the parsimonious model is chosen based on the corrected Akaike's Information Criterion (AICc) (Sugiura, 1978) that is better suited for model selection in small samples than AIC:

$$\text{AICc} = N \ln \text{RMSE}^2 + 2K + \frac{2K(K+1)}{N-K-1}$$

where *N* is the number of samples and *K* the number of factors. The model that represents an 'elbow' in the scree plot (AICc versus number of factors curve) is chosen. Stone (1977) showed that leave-one-out cross-validation and AIC were asymptotically equivalent. In small samples AIC overfits suggesting that leave-one-out cross-validation might have the same property. Burnham and Anderson (2002) describe model selection strategies in biological sciences based on AIC and related criteria and advise using the AICc. Some further measures to assess the goodness of the model are the bias (BIAS):

$$\text{BIAS} = \frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)$$

the standard deviation of the error distribution (SDE):

$$\text{SDE} = \sqrt{\frac{\sum_{i=1}^N ((\hat{y}_i - y_i) - \text{BIAS})^2}{N-1}}$$

and the residual predictive deviation (RPD) – the ratio of the standard deviation of the data to the RMSE.

3. Results and discussion

3.1. Calibration and validation

The diffuse spectra show a maximum of absorption at about 630 nm which corresponds to the Brilliant Blue absorption maximum in water (e.g. Forrer, 1997). The second smaller peak of the Brilliant Blue absorption spectrum is situated at about 430 nm (Fig. 1). While the concentration of the dye is similar for all samples (Fig. 1), the shapes of the spectra differ. Especially the horizons Ea, Bsh and Bs show higher absorbance between 400 and 430 nm indicating that

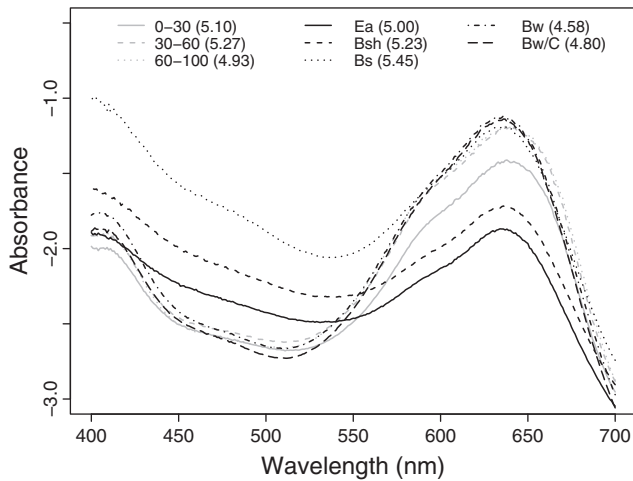


Fig. 1. Soil visible diffuse spectra. The content of Brilliant Blue in mg per g soil is indicated in parenthesis.

other parameters than Brilliant Blue concentration influence the spectra of stained samples in the visible range.

The leave-one-out cross validation suggested a model with four factors for calibration set I and a model with six factors for calibration set II (Fig. 2). The first four factors account for 95.61% of variation in the Brilliant Blue concentrations and 99.99% of variation in the spectra of calibration set I. In the calibration set II, the model explained 95.17% of variation in the Brilliant Blue concentrations and 99.99% of variation in the spectra. The leave-one-out cross validation statistics is similar for the two calibration sets. The RMSE of about 1 mg g^{-1} and an R^2_{adj} of 0.94 indicate good models (Table 2). The fitted versus observed values plots did not show any non linearity (Fig. 3a and b). In the calibration set I, one point with a high residual was detected, but it did not deteriorate the RMSE of the validation.

The statistics of the leave-one-out cross validation and the validation based on bagging-PLSR are comparable for the agricultural site, but differ in RPD for the forest site (Table 3). However, a lower

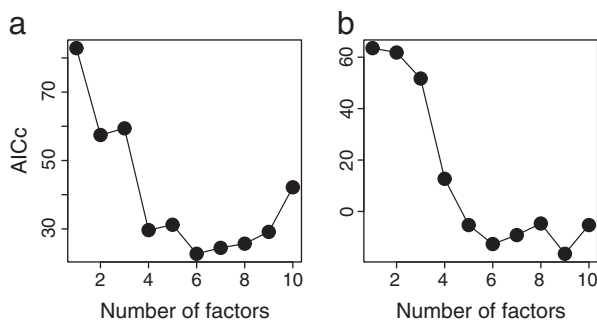


Fig. 2. Evolution of the corrected Akaike's Information Criterion (AICc) with increasing number of factors; (a) calibration set I and (b) calibration set II.

Table 2
Statistics of the calibration procedure based on leave-one-out cross validation.

Calibration set	RMSE ^a (mg g^{-1})	R^2_{adj}	BIAS (mg g^{-1})	SDE ^b (mg g^{-1})	RPD ^c
I	1.29	0.94	−0.01	1.31	4.04
II	0.82	0.94	−0.01	0.82	4.10

^a Root mean square error.

^b Standard deviation of the error distribution.

^c Ratio of the standard deviation of the data to the RMSE.

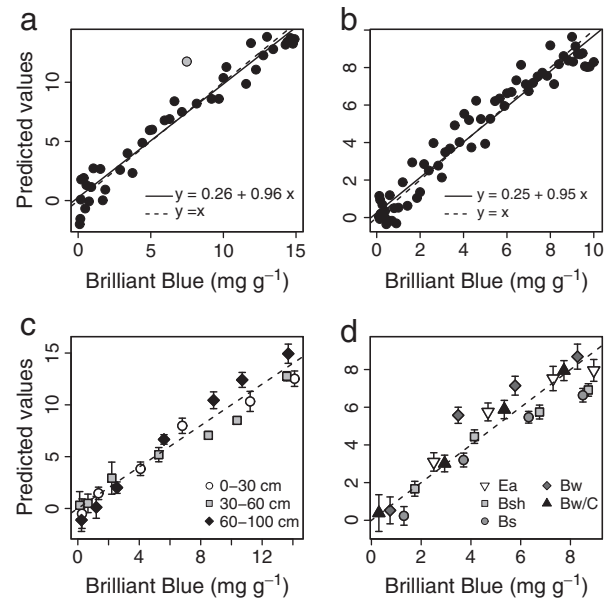


Fig. 3. Results of the leave-one-out cross validation: (a) calibration set I (the grey point has an unusually high residual) and (b) calibration set II; and of the validation procedure based on bagging-PLSR: (c) validation set I and (d) validation set II.

Table 3
Statistics of the validation procedure based on bagging-PLSR.

Validation set	RMSE ^a (mg g^{-1})	R^2_{adj}	BIAS (mg g^{-1})	SDE ^b (mg g^{-1})	RPD ^c
I	1.09 (0.83, 1.59) ^d	0.95	−0.16	1.11	4.57
II	0.98 (0.75, 1.44) ^d	0.87	−0.08	1.00	2.87
I and II	1.35 (1.10, 1.73) ^d	0.88	0.14	1.36	3.00

^a Root mean square error.

^b Standard deviation of the error distribution.

^c Ratio of the standard deviation of the data to the RMSE.

^d 95% confidence interval.

RPD in the validation is due to a smaller standard deviation of concentrations in the validation set II and not to a degradation of the RMSE. Validation set I gave an RMSE of 1.09 mg g^{-1} and an R^2_{adj} of 0.95. For validation of set II we obtained an RMSE of 0.98 mg g^{-1} and an R^2_{adj} of 0.87. All depths in validation set I and horizons in validation set II were equally well predicted (Fig. 3c and d).

If the main goal in estimating Brilliant Blue concentrations is to correct C and N contents in soil in a dye tracer study, the accuracy of prediction should be comparable to the accuracy of a standard CN-analyser. An error of 0.1% for C and 0.01% for N should be acceptable and equals an RMSE of 1.78 mg of Brilliant Blue per g soil. As the RMSE of our models are smaller they can be used to correct C and N contents.

3.2. Robustness of the PLSR models

To infer the robustness of the models, we predicted the concentrations of Brilliant Blue in validation set I using the calibration set II and vice versa. Using the same PLSR models as described above yielded an R^2_{adj} of 0.22, an RMSE of 13.52 mg g^{-1} and a BIAS of $−10.65 \text{ mg g}^{-1}$ for validation set II. Prediction for calibration set I based on calibration set II was better, leading an R^2_{adj} of 0.83, an RMSE of 3.71 mg g^{-1} and a BIAS of 2.86 mg g^{-1} . We attribute this deterioration of prediction ability to variations in soil colour, as the concentration range of Brilliant Blue is similar in both calibration sets. Soil colour must be considered as a nuisance parameter that changes the hue of samples with the same concentration of Brilliant Blue, thus

affecting their vis diffuse spectra. Therefore, a calibration procedure should take not only concentrations of Brilliant Blue, but also soil background colours into account.

To improve prediction we combined both calibration sets and built a new model. The pre-processing of the spectra had to be adjusted to account for a more heterogeneous data set. In addition to wavelet smoothing and mean-centring we applied a wavelet detrending and calculated the first derivative. Wavelet detrending is often used to correct baseline shifts or to remove curvilinearity in the spectra (Viscarra Rossel et al., 2007). For the combined data set, only 5 factors were necessary. The prediction of both validation sets together was satisfactory and gave an R^2_{adj} of 0.88 and an RMSE of 1.35 mg g^{-1} (Table 3). The model explained 90.18% of variation in Brilliant Blue concentrations and 98.13% of variation in the spectra.

4. Conclusions

Rapid and accurate estimation of Brilliant Blue by vis DRS is possible. However, the accuracy of prediction can be altered when soil background colour varies between calibration and prediction data sets. The hue of Brilliant Blue stained soil depends on the soil background color and the amount of the dye. From the failure to predict the validation set II using the calibration set I and vice versa we conclude that not only the range of concentrations, but also the soil color should be considered when calibrating the model. Therefore, we recommend a site specific calibration based on typical soil colours and an expected range of Brilliant Blue concentrations.

To deal with varying soil colours, NIR (near infrared) spectra of the soil could be used additionally to the visible range. Indeed, soil colour is strongly affected by the content of organic matter and iron and aluminium oxides and hydroxides. Several authors (Islam et al., 2003; Viscarra Rossel et al., 2006) showed that organic C, iron and aluminium could be estimated by NIR DRS. Further studies should investigate whether taking the NIR range into account can improve the robustness of the PLSR models.

Acknowledgements

The first author was financially supported by the Deutsche Forschungsgemeinschaft (DFG FOR 562) and the Deutsch-Französische

Hochschule (Grant no. CT-31-06). We thank the anonymous reviewers for helpful comments on the manuscript.

References

- Bundt, M., Jaggi, M., Blaser, P., Siegwolf, R., Hagedorn, F., 2001a. Carbon and nitrogen dynamics in preferential flow paths and matrix of a forest soil. *Soil Science Society of America Journal* 65, 1529–1538.
- Bundt, M., Widmer, F., Pesaro, M., Zeyer, J., Blaser, P., 2001b. Preferential flow paths: biological 'hot spots' in soils. *Soil Biology & Biochemistry* 33, 729–738.
- Burnham, K., Anderson, D., 2002. Model selection and multimodel inference: a practical information-theoretic approach. Springer Verlag, New York.
- Flury, M., Flühler, H., Jury, W.A., Leuenberger, J., 1994. Susceptibility of soils to preferential flow of water: a field study. *Water Resources Research* 30, 1945–1954.
- Forrer, I.E., 1997. Solute transport in an unsaturated field soil: visualisation and quantification of flow patterns using image analysis. Ph.D. thesis. Swiss Federal Institute of Technology.
- Forrer, I.E., Papritz, A., Kasteel, R., Flühler, H., Luca, D., 2000. Quantifying dye tracers in soil profiles by image processing. *European Journal of Soil Science* 51, 313–322.
- Geladi, P., Kowalski, B.R., 1986. Partial least-squares regression – a tutorial. *Analytica Chimica Acta* 185, 1–17.
- Hagedorn, F., Mohn, J., Schleppei, P., Flühler, H., 1999. The role of rapid flow paths for nitrogen transformation in a forest soil: a field study with micro suction cups. *Soil Science Society of America Journal* 63, 1915–1923.
- Islam, K., Singh, B., McBratney, A., 2003. Simultaneous estimation of several soil properties by ultra-violet, visible, and near-infrared reflectance spectroscopy. *Australian Journal of Soil Research* 41, 1101–1114.
- IUSS Working Group WRB, 2007. World reference base for soil resources 2006. first update 2007 World Soil Resources Reports No. 103. <http://www.fao.org/ag/agl/agll/wrb/doc/wrb2007>. [Accessed on 21-June-2010].
- Oyama, M., Takehara, H., 1999. Revised standard soil color charts. Eijkkelkamp Agrisearch Equipment. Soil colour book 08.11, Netherlands.
- Stone, M., 1977. An asymptotic equivalence of choice of model by cross-validation and akaike's criterion. *Journal of the Royal Statistical Society. Series B (Methodological)* 39, 44–47.
- Sugiura, N., 1978. Further analysts of the data by akaike's information criterion and the finite corrections. *Communications in Statistics – Theory and Methods* 7, 13–26.
- Viscarra Rossel, R.A., 2007. Robust modelling of soil diffuse reflectance spectra by "bagging-partial least squares regression". *Journal of Near Infrared Spectroscopy* 15, 39–47.
- Viscarra Rossel, R.A., 2008. Parles: Software for chemometric analysis of spectroscopic data. *Chemometrics and Intelligent Laboratory Systems* 90, 72–83.
- Viscarra Rossel, R.A., Walvoort, D.J.J., McBratney, A.B., Janik, L.J., Skjemstad, J.O., 2006. Visible, near infrared, mid infrared or combined diffuse reflectance spectroscopy for simultaneous assessment of various soil properties. *Geoderma* 131, 59–75.
- Viscarra Rossel, R.A., Taylor, H.J., McBratney, A.B., 2007. Multivariate calibration of hyperspectral γ -ray energy spectra for proximal soil sensing. *European Journal of Soil Science* 58, 343–353. doi:10.1111/j.1365-2389.2006.00859.x.
- Wehrens, R., Putter, H., Buydens, L.M.C., 2000. The bootstrap: a tutorial. *Chemometrics and Intelligent Laboratory Systems* 54, 35–52.